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Beyond a single structure: complementary IR and NMR signatures of multi-minima noncovalent binding

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Hydridic hydrogen atoms, where hydrogen carries increased electron density, form a distinct class of noncovalent interactions, yet their structural and energetic characteristics remain poorly understood. Shallow multi-minima potential-energy surfaces and competing low-energy docking motifs make conventional NMR insufficient to resolve individual interactions. Using a triethylsilane...1,4-diiodoperfluorobutane complex, we demonstrate that hydridic Si–H...I interactions coexist with competing C–H...I, C–H...F and repulsion-wall motifs as well as numerous dispersion energy contributions producing diverse Si–H vibrational signatures in cryogenic ATR-IR spectra. Combined with quantum-chemical calculations, these data reveal that hydridic hydrogen bonds stabilize specific docking geometries, modulate local electrostatics, and contribute to association energies, even when conventional NMR signals appear averaged. Our results establish hydridic hydrogen bonding as a tunable, structurally and energetically distinct noncovalent interaction, providing a framework for rational design of organosilicon catalysts, supramolecular assemblies, and molecular recognition systems in chemistry and biology.

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Introduction

Non-covalent interactions—including hydrogen and halogen bonding, σ - and π -hole interactions, charge-transfer, dispersion, and related electrostatic and induction effects—often govern molecular recognition, self-assembly, and reactivity.^{1–8} Hydrogen's role in noncovalent chemistry is usually framed in terms of a protonic donor. In the classical hydrogen bond (X–H...Y), a polarized X–H group presents a region of positive electrostatic potential that binds to a negative region (Y) possessing lone pair or π -density, and charge transfer is directed from Y to the X–H σ^* orbital.^{9–11} Yet hydrogen can also occupy the opposite end of this polarity spectrum. When bound to electropositive elements, hydrogen may become hydridic—bearing increased electron density and acting as a nucleophilic site.^{12,13} In this case, X–H serves as electron donor, not an

electron acceptor, and charge transfer proceeds from hydridic hydrogen to an electron acceptor, *i.e.* similar to the case of a halogen bond. This inverted polarity is central to hydride transfer and catalytic bond activation,¹⁴ but it also opens a less-charted regime of noncovalent interactions whose structural and energetic characteristics are poorly understood – particularly, how hydridic hydrogen engages electrophilic partners through electrostatics, induction, and dispersion.^{15,16}

Because hydrogen is light and highly polarizable, it functions as a sensitive probe of weak intermolecular forces: small changes in local electrostatics, charge transfer, dispersion, and steric compression can produce measurable spectroscopic responses.^{17–19} In IR, the X–H stretching vibration responds directly to perturbations of bond strength and length, leading to red shifts for bond weakening and blue shifts for compressive contacts,^{20–23} with empirical links between stretching shifts and hydrogen bonding geometry established early on.²⁴ In parallel, ¹H NMR chemical shifts and their temperature/concentration dependences provide an atomic-scale probe of hydrogen's electronic environment and dynamics, but—precisely because they are ensemble-averaged—can conceal competing motifs under fast exchange, even as they remain powerful reporters when combined with complementary observables.^{25,26} This combination makes hydrogen-centered spectroscopy particularly useful for disentangling multi-minima noncovalent landscapes.

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At the same time, applying an σ -hole (“halogen bond”) or π -hole contacts whose dominant experimental observables are hydrogen-centered can be conceptually counterproductive. In practice, these assemblies are mostly detected directly and quantified through perturbations of the X–H bond—*via* shifts, splittings and intensity changes of the X–H stretching vibration in IR—and through changes in ^1H shielding and its temperature/concentration dependence in NMR.^{27,28} These readouts report how local electrostatics, induction, and steric compression act on hydrogen and the X–H bond.^{29,30} Describing such interactions primarily as halogen bonds because the electrophilic site is a σ -hole on Hal-Z risks an unintuitive inversion: the interaction is classified based on the acceptor fragment but measured based on hydrogen-based reporters, so a contact distinguished by an X–H stretching shift and a ^1H chemical-shift response is declared “not a hydrogen bond.” This tension is not merely semantic: it can obscure experimentally accessible structure–spectra relationships and encourage overinterpretation of averaged NMR shifts in multi-minima landscapes. A more useful description should therefore retain the hydrogen-centred perspective—recognizing X–H $\cdots$ (σ/π -hole) contacts as a hydridic hydrogen bond regime—while still acknowledging that σ -hole donation by Hal-Z is an important part of the driving physics. Also note that the σ -hole model has been critically discussed, particularly with respect to its purely electrostatic interpretation, and that covalent/orbital-interaction contributions may be essential for a complete description.³¹

Although the sign of the hydrogen partial charge is inverted and the total charge-transfer direction is reversed, the vibrational response upon complexation is almost identical to that of conventional (protonic) hydrogen bonds.^{32,33} For this reason, simple, well-defined model systems are invaluable for connecting IR and ^1H NMR observables to specific interaction motifs and for testing bond-type assignments in the presence of competing docking geometries.

In the present study we use the triethyl silane (Et_3SiH) $\cdots$ 1,4-diiodoperfluorobutane ($\text{C}_4\text{F}_8\text{I}_2$) pair as a genuine multi-minima system at the interface of hydridic-hydrogen and σ -hole chemistry. Binding in the $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ complex can be characterized by either a Si–H $\cdots$ I hydridic hydrogen bond or H $\cdots$ I–C halogen bond. Following the IUPAC convention the interaction is named after the electron-accepting site; therefore, it should be classified as a halogen bond. The problem remains the experimental detection of both bonding motifs which is straightforward for the hydrogen bond but more involved for the halogen bond (see later). Et_3SiH enables competition between hydrogen-centered motifs, while each iodine atom in $\text{C}_4\text{F}_8\text{I}_2$ features a positive σ -hole along the C–I bond extension and a negative equatorial belt, with additional negative regions on fluorine atoms. We apply systematic conformational and docking sampling to resolve competing interaction motifs and experimentally observed Si–H perturbations as an ensemble of low-lying minima rather than a single structure. Silicon hydrides provide a particularly sensitive probe of weak association because the Si–H bond responds strongly to subtle changes in local electrostatics and polarization.^{34,35} Our

experimental and theoretical results reveal that hydridic hydrogen bonds exhibit a rich structural and energetic diversity: distinct low-energy docking motifs coexist, each producing characteristic spectroscopic signatures in Si–H vibrations. Importantly, findings show that hydridic hydrogen bonds can stabilize specific molecular arrangements and modulate local electrostatics, providing a tunable interaction for organosilicon catalysis, supramolecular assembly, and molecular recognition.

One reason for selecting the present complex was a recent experimental ^{19}F NMR study,³⁶ in which the interaction was interpreted as a halogen bond. However, that analysis considered only a single optimized structure and did not account for the multimimum character of the potential-energy surface. Consequently, the experimental NMR observables were assigned to one minimum rather than interpreted as Boltzmann-averaged properties of an ensemble, and the selected structure did not correspond to the global minimum. A central aim of the present work is therefore to demonstrate how such flexible noncovalent complexes, characterized by several closely lying minima, should be analysed and related to ensemble-averaged experimental observables. Consequently, the use of the NMR technique for this and similar complexes is limited.

Experimental

NMR spectroscopy

Six samples were prepared: (1) Et_3SiH (17 μL) in deuterated hexane (C_6D_{14} , 550 μL), (2) $\text{C}_4\text{F}_8\text{I}_2$ (20 μL) in C_6D_{14} (550 μL), (3) Et_3SiH (17 μL) in $\text{C}_4\text{F}_8\text{I}_2$ (550 μL), (4) $\text{C}_4\text{F}_8\text{I}_2$ (20 μL) in Et_3SiH (550 μL), (5) an equimolar mixture of Et_3SiH (17 μL) and $\text{C}_4\text{F}_8\text{I}_2$ (20 μL) in C_6D_{14} , and (6) a mixture of Et_3SiH and $\text{C}_4\text{F}_8\text{I}_2$ (1 : 40 v/v, 316 μL) in C_6D_{14} . The NMR spectra were measured on a 500 MHz Bruker Neo spectrometer (^1H at 500.3 MHz, ^{19}F at 470.7 MHz). The sample temperature was calibrated using methanol. The ^1H chemical shifts were referenced to the methyl signal of Et_3SiH ($\delta = 1.049$ ppm). The ^{19}F chemical shifts were referenced to the $\text{CF}_2\text{--CF}_2$ signal of $\text{ICF}_2\text{--CF}_2\text{--CF}_2\text{--CF}_2\text{I}$ ($\delta = -109$ ppm).

IR spectroscopy

Cryogenic ATR-IR measurements were performed using a Bruker Vertex 70v FT-IR spectrometer equipped with a liquid-nitrogen-cooled MCT detector, and a Specac Low Temperature Golden Gate™ diamond ATR accessory with a high-stability temperature controller. Low-temperature experiments were carried out under vacuum, and the optical bench was simultaneously maintained under full evacuation to suppress atmospheric water and CO_2 contributions. Spectra were collected at a spectral resolution of 0.5 cm^{-1} by averaging 128 scans to improve the signal-to-noise ratio. Liquid samples (10 μL) were deposited onto the ATR crystal at the target temperature before spectral acquisition. The $\text{Et}_3\text{SiH}/\text{C}_4\text{F}_8\text{I}_2$ mixture was prepared at a 1 : 40 (v/v) volume ratio, while measurements in *n*-hexane were carried out at concentrations of 2.8 M for $\text{C}_4\text{F}_8\text{I}_2$ and 0.08 M for Et_3SiH . To assess the reversibility of complex formation, the sample was applied to the ATR crystal after the system had

reached the desired cryogenic temperature, and independent spectra were recorded immediately. The background spectra were recorded at the same set temperature in advance.

IR transmittance spectra were analyzed in OriginPro (OriginLab Corporation) to determine the full width at half maximum (FWHM) of the absorption bands. Peak fitting was performed using the built-in PsdVoigt1 function from the OriginPro fitting library, which represents a pseudo-Voigt line shape as a weighted combination of Lorentzian and Gaussian contributions:

$$y(x) = y_0 + A[\mu L(x) + (1 - \mu)G(x)]$$

with

$$L(x) = \frac{2}{\pi} \times \frac{w}{4(x - x_c)^2 + w^2}$$

and

$$G(x) = \frac{\sqrt{4 \ln(2)}}{\sqrt{\pi}w} \exp\left[-\frac{4 \ln(2)}{w^2}(x - x_c)^2\right]$$

Here, y_0 is the baseline offset, A is the peak amplitude, x_c is the band center, w is the FWHM, and μ is the Gaussian–Lorentzian mixing parameter. The fitting procedure employed nonlinear least-squares minimization, with initial estimates of the fitting parameters generated automatically and subsequently refined iteratively until convergence. The quality of the fit was evaluated using the coefficient of determination (R^2) and adjusted R^2 . The FWHM values were taken directly from the fitted parameter w , enabling accurate determination of band widths with reduced sensitivity to spectral noise. The corresponding pseudo-Voigt fitting parameters for the spectra recorded at -40 °C and -10 °C are summarized in Table S1.

Calculations

Geometry optimization and selection of complexes: initial $\text{Et}_3\text{SiH} \cdots \text{C}_4\text{F}_8\text{I}_2$ geometries were generated with in-house Python scripts written as XYZ files. Two complementary placement protocols were employed:

(1) Iodine-targeted sampling around the Si–H bond: for each trial complex, one of the two iodine atoms of the guest fragment was selected and placed at a prescribed starting distance $r(\text{I} \cdots \text{H}-\text{Si})$ from the hydride hydrogen of Et_3SiH . The position of iodine was defined in spherical coordinates around the $\text{Si} \rightarrow \text{H}$ axis, where $\theta = 0^\circ$ corresponds to a collinear $\text{I} \cdots \text{H}-\text{Si}$ arrangement. In the generator, $r(\text{I} \cdots \text{H}-\text{Si})$ was sampled in the range 2.6–4.0 Å, θ in 0° – 60° , and the azimuthal angle ϕ in 0° – 360° . The guest fragment was additionally given a random overall rotation before translation.

(2) Systematic C–H $\cdots$ I sampling opposite to the Si–H bond (“bottom-site” protocol): a “bottom” C–H site on Et_3SiH was chosen as the C–H hydrogen whose direction from Si is maximally opposed (largest angle) to the Si–H hydride direction. The guest fragment was then oriented so that a chosen C–I bond was approximately collinear with the extension of that C–H bond, and the $\text{X} \cdots \text{H}$ distance (here $\text{X} = \text{I}$) was scanned over a user-

defined grid (multiple distances and azimuthal rotations about the $\text{H} \cdots \text{I}$ axis). To enforce that the interaction involves the selected C–H site (and not the hydride), structures were rejected if the Si–H hydride hydrogen approached any guest atom closer than a preset exclusion distance.

For both protocols, candidate structures were filtered to remove severe inter-fragment overlaps by rejecting trials where the minimum host–guest atom–atom distance fell below a threshold (e.g., 2.0 Å in the targeted generator).

Every generated complex was geometry-optimized using the L-BFGS optimizer of ASE with energies and forces provided by the UMA-S-1P1 universal machine-learning interatomic potential accessed through the FAIRChemCalculator (fairchem-core).³⁷ Calculations were run in the molecular task mode (task_name=“omol”), as recommended for molecular chemistry use cases. In our workflow, the spin multiplicity was supplied through the ASE Atoms metadata (atoms.info[“spin”]), and singlet complexes were treated with spin = 1 as used in the structure generation scripts. After UMA optimization, the selected most reliable geometries were reoptimized using the ORCA 6.0.1 (ref. 38 and 39) software package at the $\omega\text{B97M-V}^{40}/\text{def2-TZVP}^{41}$ level. After ORCA reoptimization, only a few geometries with the lowest energy and different structures were selected.

Final geometry optimizations were performed at the $\omega\text{B97M-V}/\text{def2-QZVP}^{41}$ level. $\omega\text{B97M-V}$ was chosen because of its excellent overall performance in comprehensive benchmark study⁴² and the good agreement with experiments observed in our previous studies.^{33,43} Optimizations were considered converged when the maximum component of the nuclear gradient fell below 5.0×10^{-5} a.u. and the total energy change between consecutive iterations was below 1.0×10^{-7} a.u. The effective core potential associated with the Karlsruhe def2 basis sets was applied to iodine. Numerical integration in the DFT calculations employed a (99, 590) grid (99 radial shells with 590 Lebedev points per shell) for the exchange–correlation functional, while the SG-1 grid was used for the VV10 nonlocal correlation term. Finally, seven low-energy complexes with no close structural similarity were selected for further analysis. Harmonic frequency calculations were then performed for each optimized structure to obtain IR frequencies and intensities. Geometry optimizations and frequency calculations were carried out in Q-Chem 6.2.⁴⁴

The shifts in the Si–H stretching frequencies were calculated as the differences between the frequencies of the complexes and the corresponding monomer frequencies. Because the complexes contain Et_3SiH monomers in different conformations, the appropriate monomer was used as the reference in each case. Specifically, monomer S_b was used for complexes **A**, **E**, and **G**; monomer S_a for complexes **B**, **C**, and **D**; and monomer S_c for complex **F**. No scaling factor was applied. The same procedure was used for the intensities, which are reported as the complex/monomer ratio.

In the Rigid Rotor Harmonic Oscillator (RRHO) framework, thermal contributions are dominated by low-frequency modes that are strongly affected by internal rotations, intermolecular vibrations, and anharmonicity; as a result, RRHO-derived free

energies can become method-dependent artifacts rather than robust, comparable thermodynamic quantities.⁴⁵ Quasi-RRHO (qRRHO) reduces but does not eliminate this problem. In many noncovalent complexes, modes below $\sim 100\text{ cm}^{-1}$ largely reflect hindered (or near-free) torsions and soft intermolecular motions rather than harmonic vibrations; qRRHO only interpolates these contributions toward a rotor-like limit, which can bias entropies and even reorder conformers or binding motifs.⁴⁶ Moreover, correct thermodynamics for multi-torsional systems is inherently multistructural and torsion-anharmonic, so single-structure (q)RRHO models can miss substantial conformational/torsional entropy and torsion-mode coupling.⁴⁷

Because canonical CCSD(T)⁴⁸ calculations are computationally too demanding for the full $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ complexes, benchmarking was carried out on four reduced $\text{Me}_3\text{SiH}\cdots\text{ICF}_3$ model complexes that retain the essential interaction motifs. Canonical CCSD(T) and DLPNO-CCSD(T)⁴⁹ calculations were performed in ORCA 6.0.1 with the cc-pwCVQZ⁵⁰ basis set and the cc-pwCVQZ-PP⁵¹ basis set on iodine, whereas the DFT methods $\omega\text{B97M}(2)$,⁵² $\omega\text{B97M-V}$, PTPSS-D3,⁵³ TPSSH,⁵⁴ and B3LYP⁵⁵ were conducted in Q-Chem 6.2 with the def2-QZVPP basis set. Based on these benchmark results, the relative populations and stability ordering of the full complexes were assessed primarily from DLPNO-CCSD(T)/cc-pwCVQZ electronic energies calculated on $\omega\text{B97M-V}/\text{def2-QZVPP}$ geometries. The qRRHO free energies are reported only for completeness and were not used for quantitative population analysis because of the uncertainty associated with low-frequency modes in these floppy complexes.

NMR shielding constants were computed with ORCA 6.0.1 at the $\omega\text{B97M-V}/\text{x2c-QZVPPall-s}^{56}$ level using the X2C relativistic Hamiltonian to account for relativistic effects of iodine. For ^1H chemical shifts, hydridic hydrogen resonances were referenced to the methyl protons of the corresponding Et_3SiH molecule; for ^{19}F chemical shifts, the ICF_2 - fluorines were referenced to the CF_2 - fluorines of $\text{C}_4\text{F}_8\text{I}_2$.

For the calculation of partial charges, we employed the atomic dipole moment corrected Hirshfeld (ADCH) population method,⁵⁷ which has been reported to provide the best correlation with experimental data⁵⁸ and then widely used NBO,⁵⁹ Hirshfeld⁶⁰ and Mulliken⁶¹ population analyses.

Electrostatic potential calculations were performed as single-point calculations on the final optimized geometries at the $\omega\text{B97M-V}/\text{def2-QZVPP}$ level, using the same computational settings as in the geometry optimizations. ESP cube files were generated on a $150 \times 150 \times 150$ grid. The corresponding ESP surfaces were visualized in VMD 1.9.3 (ref. 62) at an electron-density isosurface of 0.001 a.u., and the mapped ESP values were refined using Multiwfn 3.8.⁶³ ALMO-EDA calculations were likewise carried out on the final optimized geometries at the $\omega\text{B97M-V}/\text{def2-QZVPP}$ level with the same integration settings using the second-generation ALMO-EDA⁶⁴ method. SCF convergence was set to 10^{-9} . The energy decomposition included permanent electrostatics, Pauli repulsion, dispersion, polarization, and charge transfer. All calculations were performed in Q-Chem 6.2.

All calculations were performed either in the gas phase or with the SMD implicit solvent model (*n*-hexane; $\epsilon = 1.83$ or triethyl silane $\epsilon = 2.32$). The solvent parameters⁶⁵ were taken as $\epsilon = 2.323$, $n_D = 1.417$, and $\gamma = 20.7\text{ mN m}^{-1}$; additionally, ϕ and ψ were set to zero because the solvent contains neither aromatic carbon atoms nor halogen atoms, and $\alpha = \beta = 0$ was assumed due to the negligible hydrogen-bond donor/acceptor character of Et_3SiH .

Results and discussion

NMR spectra

For ^1H NMR (Fig. 1a) we prepared Et_3SiH in deuterated hexane (C_6D_{14} ; dark green), an equimolar mixture of Et_3SiH and $\text{C}_4\text{F}_8\text{I}_2$ in C_6D_{14} (green), a 1 : 40 (v/v) mixture of Et_3SiH and $\text{C}_4\text{F}_8\text{I}_2$ in C_6D_{14} (light green), and Et_3SiH in $\text{C}_4\text{F}_8\text{I}_2$ (pale green). For ^{19}F NMR (Fig. 1b) we prepared $\text{C}_4\text{F}_8\text{I}_2$ in C_6D_{14} (brown), an equimolar mixture of $\text{C}_4\text{F}_8\text{I}_2$ and Et_3SiH in C_6D_{14} (orange), and $\text{C}_4\text{F}_8\text{I}_2$ in Et_3SiH (light orange). We anticipated that cooling would stabilize the $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ complex and enhance temperature-dependent chemical-shift changes.

Indeed, the ^1H chemical shifts of Si-H hydrogen (Fig. 1a) differed markedly in neat $\text{C}_4\text{F}_8\text{I}_2$ and in excess of $\text{C}_4\text{F}_8\text{I}_2$ in C_6D_{14} relative to monomeric Et_3SiH in C_6D_{14} and to the equimolar mixture in C_6D_{14} . The Si-H resonance of Et_3SiH shifted substantially upfield in excess $\text{C}_4\text{F}_8\text{I}_2$ relative to monomeric Et_3SiH in C_6D_{14} and to the equimolar mixture, and this difference became more pronounced at lower temperature. In the equimolar mixture in C_6D_{14} , the ^1H chemical shift fell between the values observed in C_6D_{14} and in $\text{C}_4\text{F}_8\text{I}_2$, consistent with partial complex formation. This upfield ^1H hydridic shift implies creation of a hydridic hydrogen bond. By contrast, the temperature dependence of the Si-H chemical shift in hexane is very small. Upon cooling, the signal shows only a slight downfield shift. This behaviour indicates that the local environment of the silane is largely temperature independent.

Comparable trends were observed in the ^{19}F spectra (Fig. 1b). The ^{19}F chemical shifts of the CF_2I groups in $\text{C}_4\text{F}_8\text{I}_2$ also differed markedly between Et_3SiH and C_6D_{14} as the solvent. With increasing temperature, the fluorine resonances shifted downfield in both solvents, but more strongly in Et_3SiH . At higher temperatures, the chemical shifts approached a common value, although the asymptote could not be reached because of solvent boiling-point constraints.

IR spectra

Fig. 2a presents the ATR-IR spectra recorded at $-140\text{ }^\circ\text{C}$. $\text{C}_4\text{F}_8\text{I}_2$ in *n*-hexane (red) shows no absorption in the $2000\text{--}2160\text{ cm}^{-1}$ region, whereas Et_3SiH in *n*-hexane (yellow) exhibits the characteristic Si-H stretching band at 2099 cm^{-1} . In the $\text{Et}_3\text{SiH}/\text{C}_4\text{F}_8\text{I}_2$ mixture (1 : 40, v/v), both in *n*-hexane and in the neat state (green and blue), an additional blue-shifted feature at 2126 cm^{-1} (blue shaded region) and two red-shifted features at 2061 cm^{-1} and 2043 cm^{-1} (red shaded region) appear alongside the Si-H stretch. Because the band positions remain unchanged upon addition of *n*-hexane, with only lower intensity and

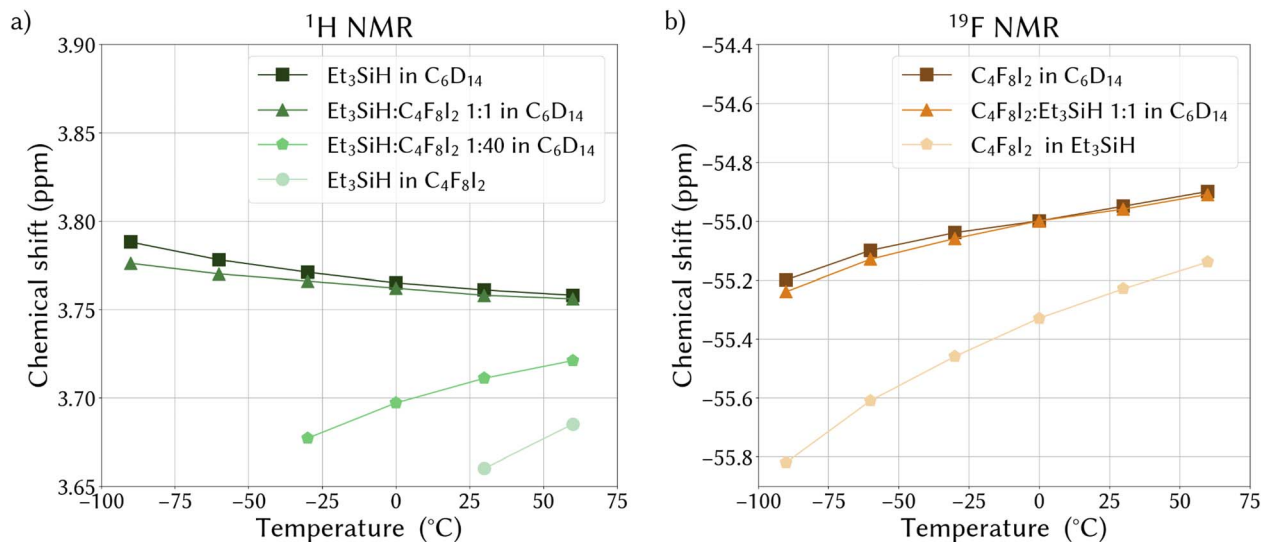


Fig. 1 The temperature dependence of (a) ^1H chemical shifts of the Si–H hydrogen atom in Et_3SiH and (b) ^{19}F chemical shifts of the ICF_2 groups of $\text{C}_4\text{F}_8\text{I}_2$. Note that the measurements of the 1 : 40 (v/v) mixture of Et_3SiH and $\text{C}_4\text{F}_8\text{I}_2$ in C_6D_{14} and Et_3SiH with $\text{C}_4\text{F}_8\text{I}_2$ used as solvent did not allow for measurements at low temperature.

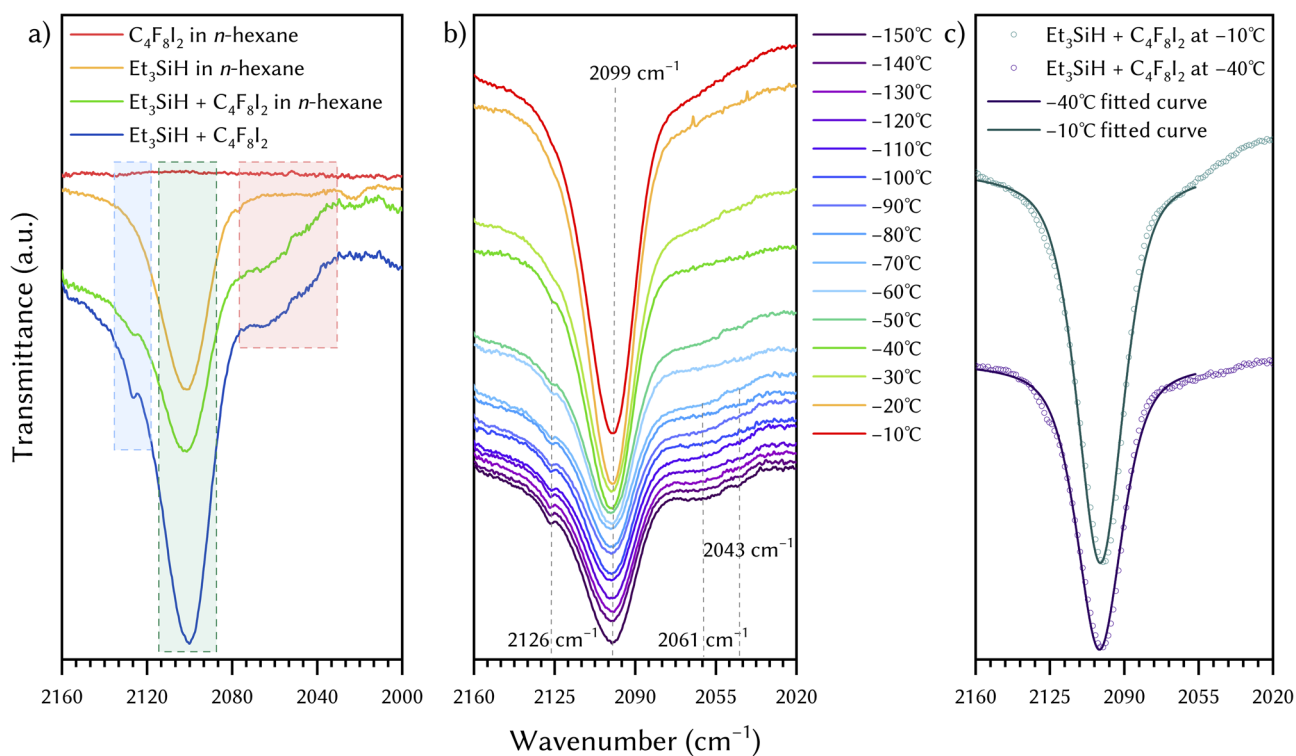


Fig. 2 (a) Cryogenic ATR-IR spectra recorded at -140°C for $\text{C}_4\text{F}_8\text{I}_2$ in n -hexane, Et_3SiH in n -hexane, and the $\text{Et}_3\text{SiH}/\text{C}_4\text{F}_8\text{I}_2$ mixture (1 : 40, v/v), measured both neat and in n -hexane solution; (b) temperature-dependent ATR-IR spectra of the $\text{Et}_3\text{SiH}/\text{C}_4\text{F}_8\text{I}_2$ mixture in the 2020–2160 cm^{-1} region recorded upon heating from -150°C to -10°C ; (c) ATR-IR spectra of the $\text{Et}_3\text{SiH}/\text{C}_4\text{F}_8\text{I}_2$ mixture recorded at -10 and -40°C in the 2020–2160 cm^{-1} region, overlaid with the corresponding pseudo-Voigt fitted curves used to determine the full width at half maximum of the absorption bands.

a reduced signal-to-noise ratio, subsequent temperature-dependent measurements were performed in the absence of a solvent.

Fig. 2b shows the temperature-dependent ATR-IR spectra of the $\text{Et}_3\text{SiH}/\text{C}_4\text{F}_8\text{I}_2$ mixture upon heating, focusing on the Si–H stretching region (2020–2160 cm^{-1}). The red-shifted bands at 2061 cm^{-1} and 2043 cm^{-1} persist up to -70°C and disappear at

higher temperatures, whereas the blue-shifted band at 2126 cm^{-1} remains detectable up to $-40\text{ }^{\circ}\text{C}$ before vanishing. Fig. 2c shows narrowing of the Si-H stretching band at 2099 cm^{-1} together with an increase in its intensity upon heating, thus supporting the existence of the two most stable structures **A** and **B** (cf. Fig. 3) even above $-40\text{ }^{\circ}\text{C}$.

In the C-H stretching region in Fig. S1 ($2800\text{--}3100\text{ cm}^{-1}$), where $\text{C}_4\text{F}_8\text{I}_2$ is featureless, the Et_3SiH band at 2950 cm^{-1} shifts by about 3 cm^{-1} in the mixture relative to the Et_3SiH monomer confirming the existence of $\text{C-H}\cdots\text{F}$ H-bonds. Together, the narrowing of the Si-H band with an increase in intensity and shifts of the C-H band provide evidence that mixing forms weakly bound complexes, consistent with the Si-H stretching region at around 2100 cm^{-1} .

Calculations

The Et_3SiH monomer contains a partially negative hydrogen bound to silicon and partially positive hydrogens on the alkyl groups (cf. Table S2). Accordingly, the ESP surface shows a negative region above the Si-H hydrogen and positive regions over the alkyl groups (Fig. S2). $\text{C}_4\text{F}_8\text{I}_2$ features electron-rich fluorine atoms and a positive σ -hole on iodine along the extension of the C-I bond, accompanied by an equatorial negative belt around the iodine atoms, as seen in the ESP surface (Fig. S3). To identify PES minima, we generated trial complexes by sampling and preoptimizing relevant monomer conformations with the Universal Model for Atoms (UMA), and then selected candidates for final geometry optimization. Details of the monomer conformers are provided in the SI.

Fig. 3 shows the seven lowest-energy minima, and Table 1 summarizes their interaction energies and key structural and

spectroscopic features. The two lowest-energy minima do not involve an $\text{Si-H}\cdots\text{X}$ contact; their stabilization arises from a network of $\text{C-H}\cdots\text{F}$ hydrogen bonds complemented by dispersion. Minima **C**, **E** and **F** exhibit an $\text{Si-H}\cdots\text{I}$ interaction in which the hydridic Si-H hydrogen engages the positive σ -hole on iodine, giving a red shift of the Si-H stretching band upon complexation. By contrast, minima **D** and **G** are dominated by $\text{C-H}\cdots\text{F}$ contacts, while a nearby fluorine atom (**D**) or iodine atom (**G**) imposes a “repulsion wall” that compresses the Si-H bond and induces a blue shift. Finally, the energy difference between energy minima is rather small, less than 0.8 kcal mol^{-1} .

The non-existence of $\text{Si-H}\cdots\text{X}$ contacts in lowest-energy minima and a small energy difference between single isomers play a key role in determining the final value of the chemical shift which is later compared with the experimental value. It is thus inevitable to confirm the existing DFT energies at a higher theoretical level, preferentially at the CCSD(T) level. Because canonical CCSD(T) is computationally prohibitive for the full $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ complexes, we benchmarked lower-cost electronic-structure protocols on reduced models that retain the key interaction motifs (four $\text{Me}_3\text{SiH}\cdots\text{ICF}_3$ complexes in Fig. S4). Table S3 summarizes the benchmark results. The best agreement with reference CCSD(T)/cc-pwCVQZ interaction energies was obtained with DLPNO/cc-pwCVQZ single points on $\omega\text{B97M-V/def2-QZVPP}$ geometries. We therefore adopted the same DLPNO workflow for the full complexes. Further details are given in the SI.

At the DLPNO level, stabilization energies are systematically smaller than the $\omega\text{B97M-V}$ ones by $\sim 0.6\text{ kcal mol}^{-1}$ and the energetic ordering changes modestly ($\omega\text{B97M-V}$: $\text{A} < \text{B} < \text{C} < \text{D} < \text{E} < \text{F} < \text{G}$; DLPNO: $\text{A} < \text{B} < \text{D} < \text{G} < \text{E} < \text{C} < \text{F}$), while the total

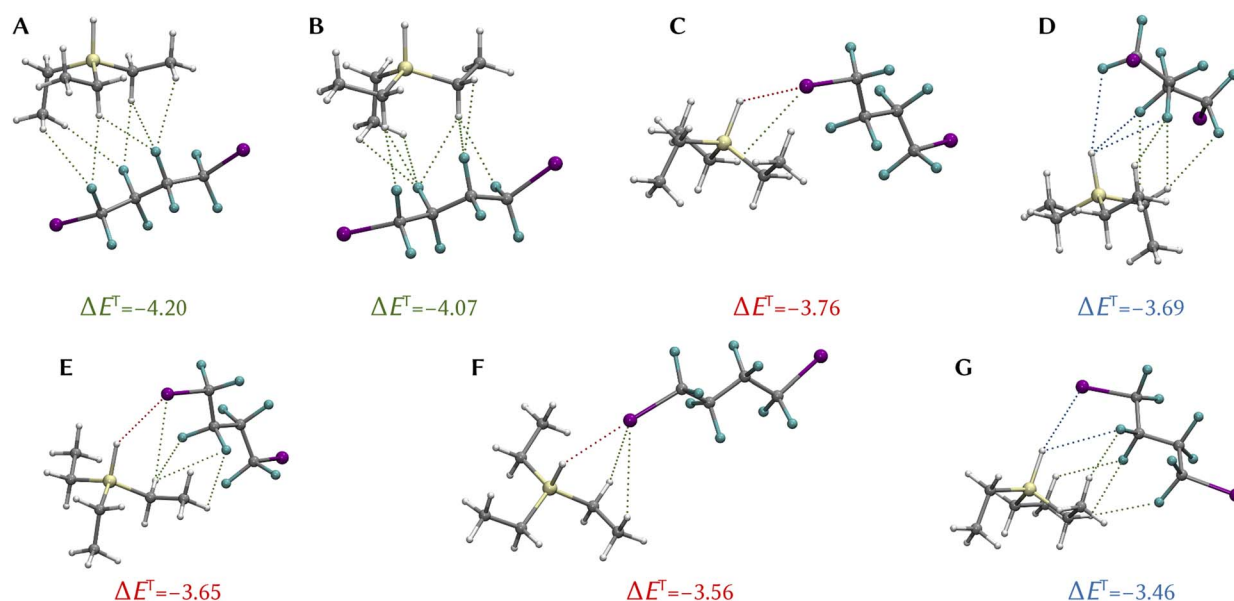


Fig. 3 Optimized geometries of $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ complexes (**A**–**G**) at the $\omega\text{B97M-V/def2-QZVPP}$ level. ΔE^{T} denotes the total interaction energy in kcal mol^{-1} for each complex. Red, blue and green colors denotes red-, blue- or no IR shift of Si-H stretching frequency upon complex formation. Dotted lines highlight contacts shorter than $3.0\text{ \AA}$ for $\text{H}\cdots\text{F}$ and $3.5\text{ \AA}$ for $\text{H}\cdots\text{I}$. Atom colors: C, gray; Si, light yellow; H, white; I, purple; F, cyan.

Table 1 Interaction energies and spectroscopic/geometric changes for $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ complexes. Total interaction energies were computed at the $\omega\text{B97M-V/def2-QZVPP}$ level ($\Delta E_{\omega\text{B97M-V}}^{\text{T}}$ in kcal mol^{-1}) on $\omega\text{B97M-V}$ geometries, and at the DLPNO/cc-pwCVQZ level (cc-pwCVQZ-PP for I; $\Delta E_{\text{DLPNO}}^{\text{T}}$ in kcal mol^{-1}) on the same geometries. Quasi-harmonic Gibbs free energies were also evaluated at the same $\omega\text{B97M-V/def2-QZVPP}$ level using the qRRHO approach ($\Delta G_{\omega\text{B97M-V}}^{123\text{K}}$ in kcal mol^{-1} at 123 K). Also shown are the change in the Si–H stretching frequency upon complexation ($\Delta\nu$ in cm^{-1}), the intensity ratio between the complex and the monomer ($I^{\text{C}}/I^{\text{M}}$), and the change in the Si–H bond length upon complexation (Δr in $\text{m}\text{\AA}$)

Complex	$\Delta E_{\omega\text{B97M-V}}^{\text{T}}$	$\Delta E_{\text{DLPNO}}^{\text{T}}$	$\Delta G_{\omega\text{B97M-V}}^{123\text{K}}$	$\Delta\nu(\text{Si-H})$	$I^{\text{C}}/I^{\text{M}}(\text{Si-H})$	$\Delta r(\text{Si-H})$
A	−4.20	−3.91	0.99	−2	1.09	0.42
B	−4.07	−3.67	0.82	−2	1.08	0.34
C	−3.76	−2.86	1.86	−50	2.03	8.79
D	−3.69	−3.28	1.21	12	0.97	−0.52
E	−3.65	−3.01	1.24	−25	1.29	4.30
F	−3.56	−2.75	1.28	−46	2.19	8.40
G	−3.46	−3.21	1.63	10	1.07	−0.18

energy window across the low-lying minima remains $\sim 0.7 \text{ kcal mol}^{-1}$. At ΔG^{123} the energy ordering changes more dramatically (**B** < **A** < **D** < **E** < **F** < **G** < **C**) but importantly, the dominant contribution of the two lowest structures remains or becomes even more pronounced. Because quasi-Rigid Rotor Harmonic Oscillator (qRRHO) has known limitations for weakly bound, floppy complexes, the present ΔG^{123} values should be interpreted with caution. In the RRHO framework, thermal contributions are dominated by low-frequency modes that are strongly affected by internal rotations, intermolecular vibrations, and anharmonicity; as a result, RRHO-derived free energies can become method-dependent artifacts rather than robust, comparable thermodynamic quantities.⁴⁵ qRRHO reduces but does not eliminate this problem. In many non-covalent complexes, modes below $\sim 100 \text{ cm}^{-1}$ largely reflect hindered (or near-free) torsions and soft intermolecular motions rather than harmonic vibrations; qRRHO only interpolates these contributions toward a rotor-like limit, which can bias entropies and even reorder conformers or binding motifs.⁴⁶ Moreover, correct thermodynamics for multi-torsional systems is inherently multistructural and torsion-anharmonic, so single-structure (q)RRHO models can miss substantial conformational/torsional entropy and torsion-mode coupling.⁴⁷

Table 1 also summarizes the changes in Si–H stretching frequency upon complexation. For structures **A** and **B**, the shift is negligible. The Si–H stretching frequency is blue-shifted in structures **D** and **G**, and red-shifted in structures **C**, **E**, and **F**. The spectral shifts correlate with changes in the Si–H bond length: bond elongation corresponds to red shifts, whereas bond shortening leads to blue shifts. Upon complexation, the Si–H stretching intensity remains essentially unchanged for the structures **A**, **B**, **D** and **G**, but increases slightly for the red-shifted ones (**C**, **E** and **F**). Si–H stretching frequencies for all monomers and complexes are summarized in Table S4.

Single isomers of the complex are stabilized by different non-covalent energy terms and the ALMO-EDA method was used for their determination (Fig. S5). Table S5 reveals a clear contrast between the red- and blue-shifted motifs (additional cases are discussed in the SI). The red-shifted complexes **C** and **F** show the strongest electrostatic and orbital stabilization (more negative E_{Elec} , E_{Pol} and E_{CT}), consistent with an $\text{Si-H}\cdots\text{I}^{\sigma\text{-hole}}$

contact, but this stabilization is offset by substantially larger Pauli repulsion. In the blue-shifted complexes, a nearby fluorine (**D**) or the negative equatorial belt of iodine (**G**) acts as a “repulsion wall” that prevents close approach to the hydridic Si–H site. This constrained, compressive Si–H contact leads to Si–H bond contraction and the observed blue shift of the stretching vibration. It also reduces short-range overlap and weakens dispersion through the increased intermolecular separation.

For inspection of the interactions between the molecules, we employed NCI analysis for all complexes **A–G** (Fig. S6 and S7). In every case, the intermolecular NCI isosurfaces are predominantly light green, indicating that the observed contacts are weakly attractive. No pronounced regions corresponding to strongly attractive interactions are present.

All reported calculations were performed with the SMD solvent model for *n*-hexane. Structures **A** and **B** lacking any Si–H $\cdots$ I contacts are predicted as the most stable ones across different computational levels. This finding is important since it indicates that conclusions about complex stability and reactivity based on IR and NMR characteristics that omit these two structures may be misleading (see later).

Resolving the spectroscopic puzzle

In the equimolar mixture, the temperature dependence of the ^1H NMR signal closely resembles that of Et_3SiH in C_6D_{14} (Fig. 1a), indicating that the complex population is too small to produce a distinct NMR signature. By contrast, excess of $\text{C}_4\text{F}_8\text{I}_2$ induces a clear upfield shift of the Si–H resonance that becomes more pronounced at lower temperature, consistent with formation of an $\text{Si-H}\cdots\text{I}^{\delta\text{-hole}}$ hydridic hydrogen bond. The ^{19}F NMR spectra (Fig. 1b) show an upfield shift upon cooling, which is more pronounced in excess Et_3SiH .

Computational sampling identifies seven low-lying minima. Their calculated ^1H NMR responses (Tables 2 and S5) differ markedly, whereas the computed ^{19}F shifts are uniformly deshielded upon complexation and remain close to those of the most stable $\text{C}_4\text{F}_8\text{I}_2$ monomer conformer (**B_{aa}**). The two lowest-energy minima, **A** and **B**, show only weak shielding of ^1H , whereas **C**, **E** and **F** are substantially more shielded, and **D** and **G** are predicted to be deshielded. In ^1H NMR, shielding of

Table 2 Calculated ^1H and ^{19}F NMR shifts (δ in ppm) for complexes of $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ calculated at the X2C- $\omega\text{B97M-V}/\text{x2c-QZVPPall-s}$ level on $\omega\text{B97M-V}/\text{def2-QZVPP}$ geometries

Monomer	$\delta(^1\text{H}; \text{Si-H})$		$\delta(^{19}\text{F}; \text{ICF}_2 \text{ group})$	
Et_3SiH (S_a)	3.015		—	
$\text{C}_4\text{F}_8\text{I}_2$ (B_{aa})	—		−55.658	
Complex	$\delta(^1\text{H}; \text{Si-H})$	$\Delta\delta(^1\text{H}; \text{Si-H})$	$\delta(^{19}\text{F}; \text{ICF}_2 \text{ group})$	$\Delta\delta(^{19}\text{F}; \text{ICF}_2 \text{ group})$
A	2.943	−0.072	−54.668	0.990
B	2.985	−0.030	−53.836	1.822
C	2.413	−0.602	−53.647	2.011
D	3.026	0.011	−54.514	1.144
E	2.854	−0.161	−54.005	1.653
F	2.328	−0.687	−53.711	1.947
G	3.359	0.344	−55.392	0.266

a hydridic X–H signal is typically associated with interaction with an electrophilic site, whereas deshielding is characteristic of interaction with a nucleophilic site. In **C**, **E**, **F** and **G**, an Si–H $\cdots$ I contact is present; it is attractive in **C**, **E** and **F**, but repulsive in **G**. Structure **D** likewise contains a repulsive Si–H $\cdots$ F contact. These repulsive contacts lead to deshielding of the Si–H signal. The experimental ^1H NMR trend is therefore compatible with some population of hydridic hydrogen bond states, but the small observed shift can also be explained by dominant contributions from minima **A** and **B**. Consistent with this interpretation, the Si–H resonance remains narrow, indicating fast exchange among a single dominant motif or several related motifs with low exchange barriers between them. By contrast, the ^{19}F response is similar across docking motifs and carries little structural information.

The temperature-dependent IR spectra (Fig. 2b) reveal two red-shifted bands and one blue-shifted band. The red-shifted bands at -38 and -56 cm^{-1} are assigned to complex **E** (calculated shift -25 cm^{-1}) and to complexes **C** and **F** (-50 and -46 cm^{-1}), respectively. The blue-shifted band at $+27\text{ cm}^{-1}$ is assigned to complexes **D** and **G**, with calculated shifts of $+12$ and $+10\text{ cm}^{-1}$. The red-shifted components remain detectable up to $-70\text{ }^\circ\text{C}$, whereas the blue-shifted component persists up to $-40\text{ }^\circ\text{C}$. From $-40\text{ }^\circ\text{C}$ to $-10\text{ }^\circ\text{C}$, the central Si–H band increases in intensity and becomes narrower (*cf.* Fig. 2c). Since the red- and blue-shifted bands have already disappeared by $-40\text{ }^\circ\text{C}$, we attribute this behaviour to dissociation of complexes **A** and **B**, whose computed Si–H shifts are negligible, and to the concomitant increase in monomeric Et_3SiH . This interpretation is consistent with the DLPNO interaction energies, which place **C**, **E** and **F** highest in energy, followed by **D** and **G**, and **A** and **B** lowest. By contrast, $\omega\text{B97M-V}$ predicts a different ordering, suggesting that the DLPNO ranking provides the more realistic description. The energy spread between the minima is only $\sim 0.8\text{ kcal mol}^{-1}$ at the $\omega\text{B97M-V}$ level and $\sim 1.2\text{ kcal mol}^{-1}$ at the DLPNO level, that is, on the scale of chemical accuracy. Additional calculations using SMD (Et_3SiH) gave qualitatively the same results; details are provided in the SI (Tables S6–S10). The IR experiments together with high-level computations

unambiguously support the existence of several complexes stabilized by the hydridic hydrogen bond showing red- or blue-shifts of the Si–H stretching frequency upon complexation.

^{19}F NMR provides no direct structural information for the present complexes, and ^1H NMR shows only that species associated with an upfield Si–H shift are populated. By contrast, IR spectroscopy reveals the multi-minima character of the system, while high-level calculations identify the low-lying structures underlying the observed spectra. Because exchange is fast on the NMR timescale, the NMR signals are narrow population-weighted averages dominated by minima **A** and **B**. The weak red- and blue-shifted IR bands indicate sparse populations of minima **C**–**G**, which explain why these species do not contribute appreciably to the observed NMR signal. NMR supports complex formation but does not resolve the detailed structure of the ensemble. This is mainly due to multi-minima character of the complex landscape, and this limitation is also valid for similar complexes with protonic and hydridic HBs possessing a multi-minima landscape. In this light, the interpretation of binding in the $\text{Et}_3\text{SiH}\cdots\text{C}_4\text{F}_8\text{I}_2$ complex as the halogen one supported by the ^{19}F NMR experiments³⁶ should be treated with caution since motifs described as halogen bonds are not expected to be directly observable by NMR when the ensemble is dominated by low-energy structures lacking specific H $\cdots$ I–C contacts.

Conclusions

This work establishes that a hydridic hydrogen bond is not merely a conceptual inversion of classical hydrogen bonds but a measurable and structurally distinct interaction that can coexist with other noncovalent motifs, such as protonic C–H $\cdots$ I, repulsion-wall and dispersion contacts. IR spectroscopy combined with quantum-chemical calculations unambiguously detects red- and blue-shifted Si–H vibrational signatures corresponding to low-energy minima, while the most stable configurations exhibit only subtle perturbations yet contribute substantially to the ensemble's stability. On the other hand, ^{19}F NMR provides only basic evidence of complex formation while

^1H NMR signals, averaged under fast exchange, primarily report the dominant minima, highlighting the multi-minima character of these complexes. NMR alone is therefore insufficient to assign a dominant bonding motif not only in the system studied but also in assemblies containing both protonic and hydridic hydrogen bonds, and thus multiple types of interactions. All these findings support describing the relevant interaction in terms of a hydridic hydrogen bond rather than a halogen bond, because the experimentally accessible signatures are centered on hydrogen rather than on iodine. Only the combination of IR and NMR spectroscopy with high-level computation resolves the full multi-minima binding landscape, with IR revealing minor minima and NMR reporting mainly dominant averaged states.

To conclude, the present study provides a framework for interpreting hydrogen-centred interactions in shallow, multi-minima landscapes and suggests that hydridic hydrogen bonds can be strategically leveraged in organosilicon catalysis, supramolecular assembly, and molecular recognition, offering a tunable handle for designing functional chemical and biochemical systems.

Author contributions

P. H., M. L., M. D., R. Z., and Š. K. designed and conceptualized the study. M. M., S. M. H. H. and Š. K. performed the IR experiment. M. D. performed the NMR experiment. M. L. performed QM calculations. P. H. and M. L. wrote the original draft of the paper. All authors discussed the results, and reviewed, edited and contributed to the final paper.

Conflicts of interest

There are no conflicts to declare.

Data availability

All data are available in the main text or the supplementary information (SI). Geometries, inputs and outputs from calculations are available *via* the Zenodo repository at <https://dx.doi.org/10.5281/zenodo.18772582>. Supplementary information: additional experimental and computational data, including ATR-IR spectra and band-fitting parameters, atomic charges and electrostatic potential surfaces, monomer conformers, interaction-energy benchmarks, Si-H stretching frequencies, ALMO-EDA and NCI analyses, and additional solvent-model calculations. See DOI: <https://doi.org/10.1039/d6sc03745c>.

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References

- 1 K. Müller-Dethlefs and P. Hobza, *Chem. Rev.*, 2000, **100**, 143–168.
- 2 A. J. Neel, M. J. Hilton, M. S. Sigman and F. D. Toste, *Nature*, 2017, **543**, 637–646.
- 3 G. Vantomme and E. W. Meijer, *Science*, 2019, **363**, 1396–1397.
- 4 R. S. J. Proctor, A. C. Colgan and R. J. Phipps, *Nat. Chem.*, 2020, **12**, 990–1004.
- 5 M. Y. Jin, Q. Zhen, D. Xiao, G. Tao, X. Xing, P. Yu and C. Xu, *Nat. Commun.*, 2022, **13**, 3276.
- 6 J. Yang, P. S. White and M. Brookhart, *J. Am. Chem. Soc.*, 2008, **130**, 17509–17518.
- 7 J. L. Fry, R. J. Rahaim Jr and R. E. Maleczka Jr, in *Encyclopedia of Reagents for Organic Synthesis*, 2007, DOI: [10.1002/9780470842898.rt226.pub2](https://doi.org/10.1002/9780470842898.rt226.pub2).
- 8 B. K. Paul, *ChemPhysChem*, 2025, **26**, e202500404.
- 9 R. H. Crabtree, P. E. M. Siegbahn, O. Eisenstein, A. L. Rheingold and T. F. Koetzle, *Acc. Chem. Res.*, 1996, **29**, 348–354.
- 10 R. W. Newberry and R. T. Raines, *Acc. Chem. Res.*, 2017, **50**, 1838–1846.
- 11 J. S. Murray and P. Politzer, *J. Indian Inst. Sci.*, 2020, **100**, 21–30.
- 12 A. Y. Houghton, J. Hurmalainen, A. Mansikkamäki, W. E. Piers and H. M. Tuononen, *Nat. Chem.*, 2014, **6**, 983–988.
- 13 N. V. Belkova, L. M. Epstein, O. A. Filippov and E. S. Shubina, *Chem. Rev.*, 2016, **116**, 8545–8587.
- 14 K. R. Brereton, N. E. Smith, N. Hazari and A. J. M. Miller, *Chem. Soc. Rev.*, 2020, **49**, 7929–7948.
- 15 S. Civiš, M. Lamanec, V. Špirko, J. Kubišta, M. Špet'ko and P. Hobza, *J. Am. Chem. Soc.*, 2023, **145**, 8550–8559.
- 16 D. W. Roberts and Y. Mao, *Phys. Chem. Chem. Phys.*, 2025, **27**, 14370–14378.
- 17 T. Seki, K.-Y. Chiang, C.-C. Yu, X. Yu, M. Okuno, J. Hunger, Y. Nagata and M. Bonn, *J. Phys. Chem. Lett.*, 2020, **11**, 8459–8469.
- 18 M. Flór, D. M. Wilkins, M. de la Puente, D. Laage, G. Cassone, A. Hassanali and S. Roke, *Science*, 2024, **386**, eads4369.
- 19 S. Tretiakov, A. Nigam and R. Pollice, *Chem. Rev.*, 2025, **125**, 5776–5829.
- 20 P. Hobza and Z. Havlas, *Theor. Chem. Acc.*, 2002, **108**, 325–334.
- 21 K. Hermansson, *J. Phys. Chem. A*, 2002, **106**, 4695–4702.
- 22 P. Hobza and Z. Havlas, *Chem. Rev.*, 2000, **100**, 4253.
- 23 E. Arunan, G. R. Desiraju, R. A. Klein, J. Sadlej, S. Scheiner, I. Alkorta, D. C. Clary, R. H. Crabtree, J. J. Dannenberg,

- P. Hobza, H. G. Kjaergaard, A. C. Legon, B. Mennucci and D. J. Nesbitt, *Pure Appl. Chem.*, 2011, **83**, 1637–1641.
- 24 G. C. Pimentel and C. H. Sederholm, *J. Chem. Phys.*, 1956, **24**, 639–641.
- 25 B. Dereka, Q. Yu, N. H. C. Lewis, W. B. Carpenter, J. M. Bowman and A. Tokmakoff, *Science*, 2021, **371**, 160–164.
- 26 L. H. E. Wieske, S. Peintner and M. Erdélyi, *Nat. Rev. Chem.*, 2023, **7**, 511–524.
- 27 V. C. Arthurs and S. A. C. McDowell, *Chem. Phys. Lett.*, 2024, **836**, 141052.
- 28 A. Mitra, P. J. Seaton, R. Ali Assarpour and T. Williamson, *Tetrahedron*, 1998, **54**, 15489–15498.
- 29 D. Manna, R. Lo, M. Lamanec, J. Pavlišová, O. Socha, M. Dračinský and P. Hobza, *J. Chem. Theory Comput.*, 2025, **21**, 7495–7502.
- 30 M. Dračinský, *Coord. Chem. Rev.*, 2026, **549**, 217284.
- 31 L. de Azevedo Santos, T. C. Ramalho, T. A. Hamlin and F. M. Bickelhaupt, *Chem.–Eur. J.*, 2023, **29**, e202203791.
- 32 M. Lamanec, J. Zienertová, M. Špejtko, D. Nachtigallová and P. Hobza, *ChemPhysChem*, 2024, **25**, e202400403.
- 33 M. Lamanec, S. Civiš and P. Hobza, *Commun. Chem.*, 2024, **7**, 254.
- 34 G. Lucovsky, *Solid State Commun.*, 1979, **29**, 571–576.
- 35 M. Uchida, T. Shimizu, R. Shibusaki, Y. Matsumoto and H. Ishikawa, *J. Chem. Phys.*, 2020, **153**, 104305.
- 36 R. M. Gomila, R. Beccaria, C. L. Iacono, A. Frontera and G. Resnati, *Chem.–Eur. J.*, 2025, **31**, e02074.
- 37 B. M. Wood, M. Dzamba, X. Fu, M. Gao, M. Shuaibi, L. Barroso-Luque, K. Abdelmaqsoud, V. Gharakhanyan, J. R. Kitchin and D. S. Levine, *arXiv*, 2025, preprint, arXiv:2506.23971, DOI: [10.48550/arXiv.2506.23971](https://doi.org/10.48550/arXiv.2506.23971).
- 38 F. Neese, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.*, 2018, **8**, e1327.
- 39 F. Neese, *Wiley Interdiscip. Rev.: Comput. Mol. Sci.*, 2022, **12**, e1606.
- 40 N. Mardirossian and M. Head-Gordon, *J. Chem. Phys.*, 2016, **144**, 214110.
- 41 F. Weigend and R. Ahlrichs, *Phys. Chem. Chem. Phys.*, 2005, **7**, 3297–3305.
- 42 N. Mardirossian and M. Head-Gordon, *Mol. Phys.*, 2017, **115**, 2315–2372.
- 43 M. Lamanec, V. Špirko, S. Civiš and P. Hobza, *J. Phys. Chem. A*, 2025, **129**, 11512–11522.
- 44 E. Epifanovsky, A. T. B. Gilbert, X. Feng, J. Lee, Y. Mao, N. Mardirossian, P. Pokhilko, A. F. White, M. P. Coons, A. L. Dempwolff, Z. Gan, D. Hait, P. R. Horn, L. D. Jacobson, I. Kaliman, J. Kussmann, A. W. Lange, K. U. Lao, D. S. Levine, J. Liu, S. C. McKenzie, A. F. Morrison, K. D. Nanda, F. Plasser, D. R. Rehn, M. L. Vidal, Z.-Q. You, Y. Zhu, B. Alam, B. J. Albrecht, A. Aldossary, E. Alguire, J. H. Andersen, V. Athavale, D. Barton, K. Begam, A. Behn, N. Bellonzi, Y. A. Bernard, E. J. Berquist, H. G. A. Burton, A. Carreras, K. Carter-Fenk, R. Chakraborty, A. D. Chien, K. D. Closser, V. Cofer-Shabica, S. Dasgupta, M. de Wergifosse, J. Deng, M. Diedenhofen, H. Do, S. Ehlert, P.-T. Fang, S. Fatehi, Q. Feng, T. Friedhoff, J. Gayvert, Q. Ge, G. Gidofalvi, M. Goldey, J. Gomes, C. E. González-Espinoza, S. Gulania, A. O. Gunina, M. W. D. Hanson-Heine, P. H. P. Harbach, A. Hauser, M. F. Herbst, M. Hernández Vera, M. Hodecker, Z. C. Holden, S. Houck, X. Huang, K. Hui, B. C. Huynh, M. Ivanov, Á. Jász, H. Ji, H. Jiang, B. Kaduk, S. Kähler, K. Khistyayev, J. Kim, G. Kis, P. Klunzinger, Z. Koczor-Benda, J. H. Koh, D. Kosenkov, L. Koulias, T. Kowalczyk, C. M. Krauter, K. Kue, A. Kunitsa, T. Kus, I. Ladjánszki, A. Landau, K. V. Lawler, D. Lefrancois, S. Lehtola, R. R. Li, Y.-P. Li, J. Liang, M. Liebenthal, H.-H. Lin, Y.-S. Lin, F. Liu, K.-Y. Liu, M. Loipersberger, A. Luenser, A. Manjanath, P. Manohar, E. Mansoor, S. F. Manzer, S.-P. Mao, A. V. Marenich, T. Markovich, S. Mason, S. A. Maurer, P. F. McLaughlin, M. F. S. J. Menger, J.-M. Mewes, S. A. Mewes, P. Morgante, J. W. Mullinax, K. J. Oosterbaan, G. Paran, A. C. Paul, S. K. Paul, F. Pavošević, Z. Pei, S. Prager, E. I. Proynov, Á. Rák, E. Ramos-Cordoba, B. Rana, A. E. Rask, A. Rettig, R. M. Richard, F. Rob, E. Rossomme, T. Scheele, M. Scheurer, M. Schneider, N. Sergueev, S. M. Sharada, W. Skomorowski, D. W. Small, C. J. Stein, Y.-C. Su, E. J. Sundstrom, Z. Tao, J. Thirman, G. J. Tornai, T. Tsuchimochi, N. M. Tubman, S. P. Veccham, O. Vydrov, J. Wenzel, J. Witte, A. Yamada, K. Yao, S. Yeganeh, S. R. Yost, A. Zech, I. Y. Zhang, X. Zhang, Y. Zhang, D. Zuev, A. Aspuru-Guzik, A. T. Bell, N. A. Besley, K. B. Bravaya, B. R. Brooks, D. Casanova, J.-D. Chai, S. Coriani, C. J. Cramer, G. Cserey, A. E. DePrinceIII, R. A. DiStasio Jr., A. Dreuw, B. D. Dunietz, T. R. Furlani, W. A. GoddardIII, S. Hammes-Schiffer, T. Head-Gordon, W. J. Hehre, C.-P. Hsu, T.-C. Jagau, Y. Jung, A. Klamt, J. Kong, D. S. Lambrecht, W. Liang, N. J. Mayhall, C. W. McCurdy, J. B. Neaton, C. Ochsenfeld, J. A. Parkhill, R. Peverati, V. A. Rassolov, Y. Shao, L. V. Slipchenko, T. Stauch, R. P. Steele, J. E. Subotnik, A. J. W. Thom, A. Tkatchenko, D. G. Truhlar, T. Van Voorhis, T. A. Wesolowski, K. B. Whaley, H. L. WoodcockIII, P. M. Zimmerman, S. Faraji, P. M. W. Gill, M. Head-Gordon, J. M. Herbert and A. I. Krylov, *J. Chem. Phys.*, 2021, **155**, 084801.
- 45 P. Pracht and S. Grimme, *Chem. Sci.*, 2021, **12**, 6551–6568.
- 46 S. Grimme, *Chem.–Eur. J.*, 2012, **18**, 9955–9964.
- 47 J. Zheng, T. Yu, E. Papajak, I. M. Alecu, S. L. Mielke and D. G. Truhlar, *Phys. Chem. Chem. Phys.*, 2011, **13**, 10885–10907.
- 48 K. Raghavachari, G. W. Trucks, J. A. Pople and M. Head-Gordon, *Chem. Phys. Lett.*, 1989, **157**, 479–483.
- 49 C. Riplinger, B. Sandhoefer, A. Hansen and F. Neese, *J. Chem. Phys.*, 2013, **139**, 134101.
- 50 K. A. Peterson and T. H. Dunning Jr., *J. Chem. Phys.*, 2002, **117**, 10548–10560.
- 51 K. A. Peterson and K. E. Yousaf, *J. Chem. Phys.*, 2010, **133**, 174116.
- 52 N. Mardirossian and M. Head-Gordon, *J. Chem. Phys.*, 2018, **148**, 241736.
- 53 L. Goerigk and S. Grimme, *J. Chem. Theory Comput.*, 2011, **7**, 291–309.

- 54 V. N. Staroverov, G. E. Scuseria, J. Tao and J. P. Perdew, *J. Chem. Phys.*, 2003, **119**, 12129–12137.
- 55 A. D. Becke, *J. Chem. Phys.*, 1993, **98**, 5648–5652.
- 56 Y. J. Franzke, L. Spiske, P. Pollak and F. Weigend, *J. Chem. Theory Comput.*, 2020, **16**, 5658–5674.
- 57 T. Lu and F. Chen, *J. Theor. Comput. Chem.*, 2012, **11**, 163–183.
- 58 S. Mahmoudi, T. Gruene, C. Schröder, K. D. Ferjaoui, E. Fröjd, A. Mozzanica, K. Takaba, A. Volkov, J. Maisriml, V. Paunović, J. A. van Bokhoven and B. K. Keppler, *Nature*, 2025, **645**, 88–94.
- 59 E. D. Glendening, C. R. Landis and F. Weinhold, *J. Comput. Chem.*, 2019, **40**, 2234–2241.
- 60 F. L. Hirshfeld, *Theor. Chim. Acta*, 1977, **44**, 129–138.
- 61 R. S. Mulliken, *J. Chem. Phys.*, 1955, **23**, 1841–1846.
- 62 W. Humphrey, A. Dalke and K. Schulten, *J. Mol. Graphics*, 1996, **14**, 33–38.
- 63 T. Lu, *J. Chem. Phys.*, 2024, **161**, 082503.
- 64 Y. Mao, P. R. Horn and M. Head-Gordon, *Phys. Chem. Chem. Phys.*, 2017, **19**, 5944–5958.
- 65 A. P. Altshuller and L. Rosenblum, *J. Am. Chem. Soc.*, 1955, **77**, 272–274.